Lean Six Sigma 4.0

Building a Continuous Improvement Mindset



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The Optimization Mandate: Creating Measurable Business Value

The objective is to identify problems, remove waste, improve processes, and create measurable business value through a synthesis of process and technology.

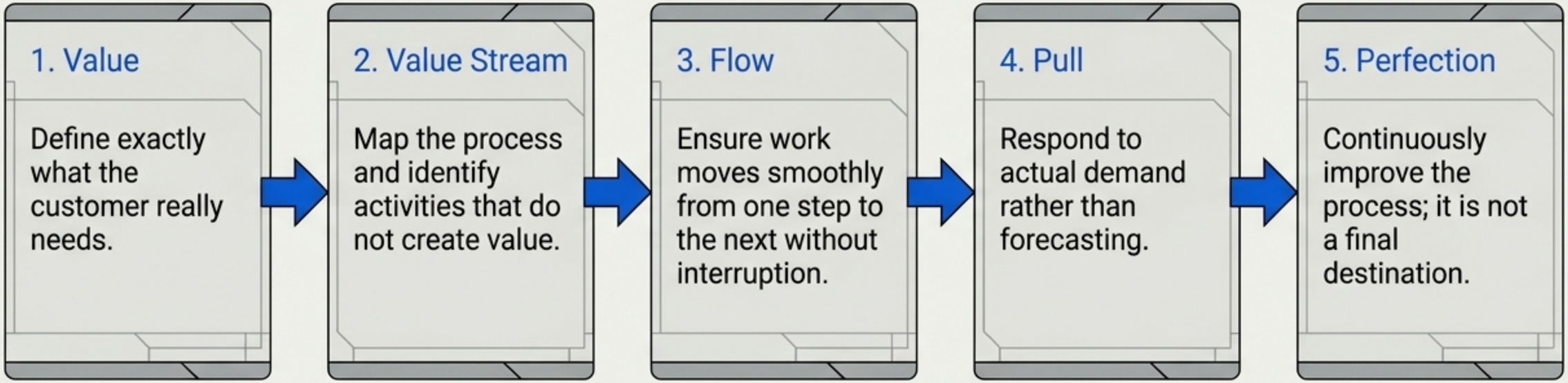
The Value Diagnostic Matrix		
Value Added (VA)	Non-Value Added but Necessary (NNVA)	Pure Waste
Directly creates value for the customer. • IT Examples: Creating a product/service, processing a customer request.	Does not create customer value but is currently required by the system.	Does not create value and must be eliminated. • IT Examples: Waiting for approval, re-entering data, fixing defects, searching for information.

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The Five Foundational Mechanics of Lean

Lean is the systematic creation of more value with less waste through customer focus, employee involvement, and process efficiency.

Process Evolution Sequence



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The IT Waste Radar: Diagnosing System Bloat

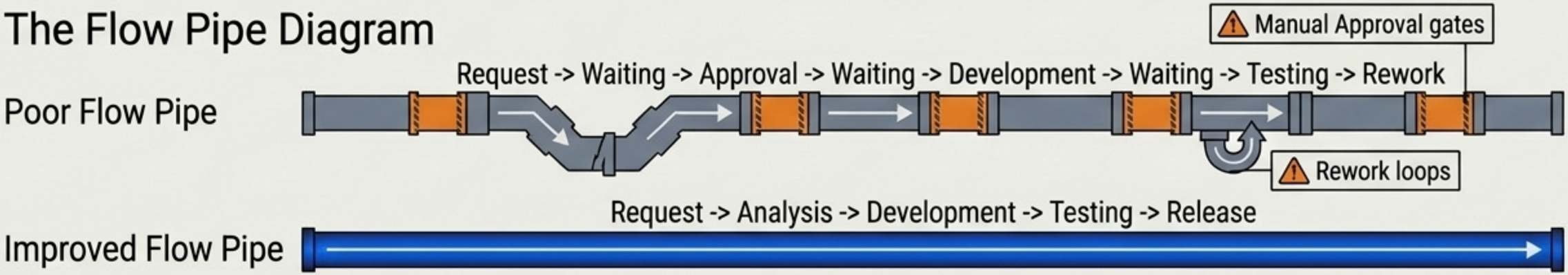
The DOWNTIME Diagnostic Grid

D Defects Production bugs, fixing repeated errors.	**O** Overproduction Building unused features based on assumptions.	**W** Waiting Waiting for manual approvals.	**N** Non-utilized talent Developer skills not fully leveraged.
T Transportation Too many teams involved in a single deployment.	**I** Inventory Duplicate data entry.	**M** Motion Searching across multiple disconnected systems.	**E** Extra-processing Unnecessary meetings and duplicated processes.

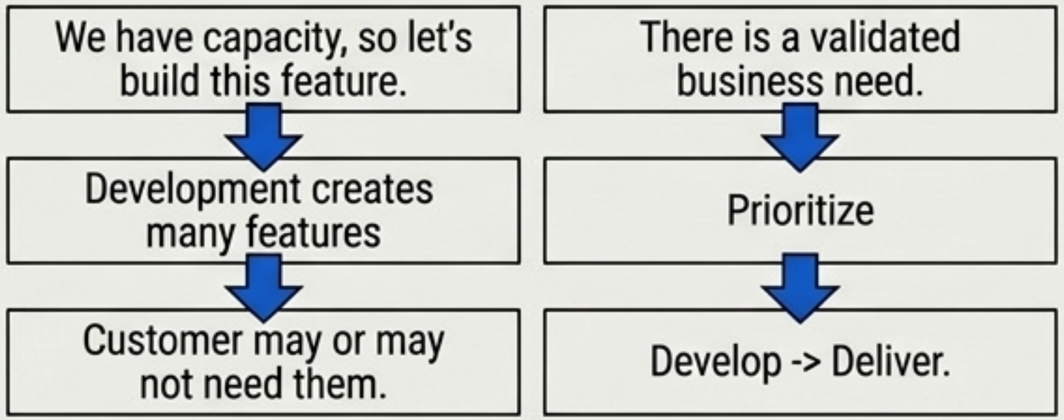
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System Architecture: Establishing Flow and Pull

The Flow Pipe Diagram



Push vs. Pull Flowchart Matrix

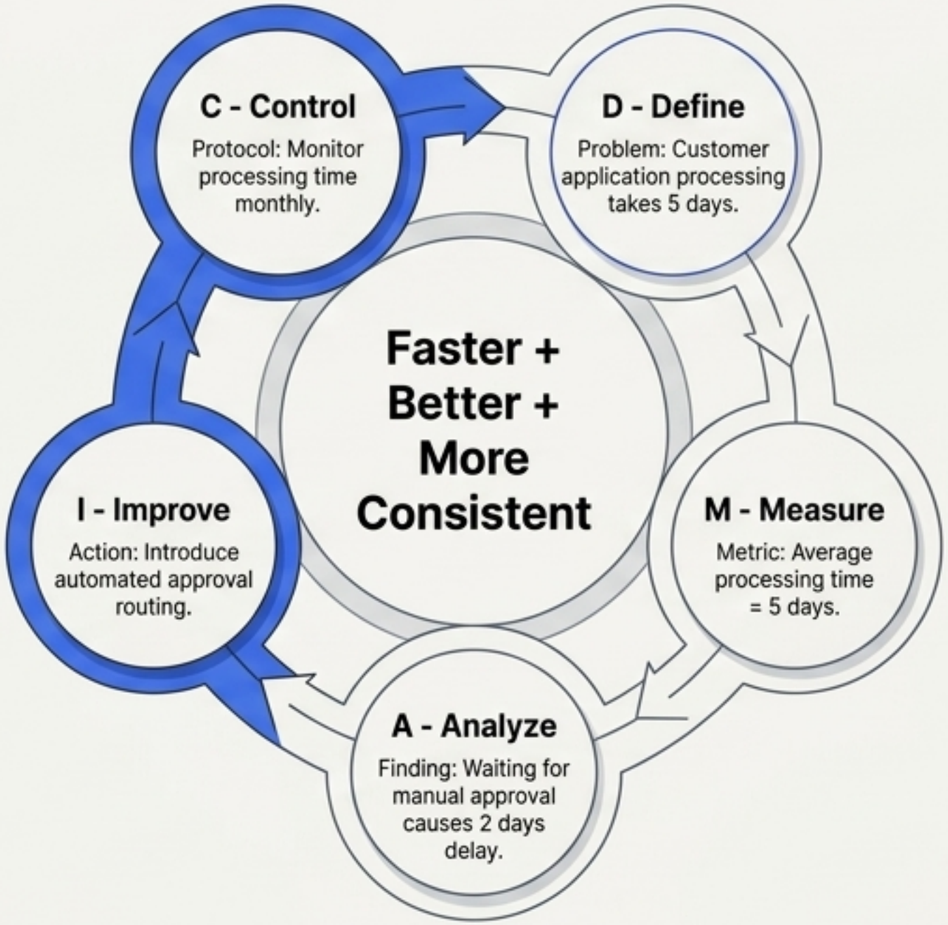


Pull System (Demand)

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The DMAIC Engine: Reducing System Variation

While Lean asks, “Where is the waste?”, Six Sigma uses data to ask, “Why is the process producing defects or variation?”



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The True Cost of Poor Quality (COPQ)

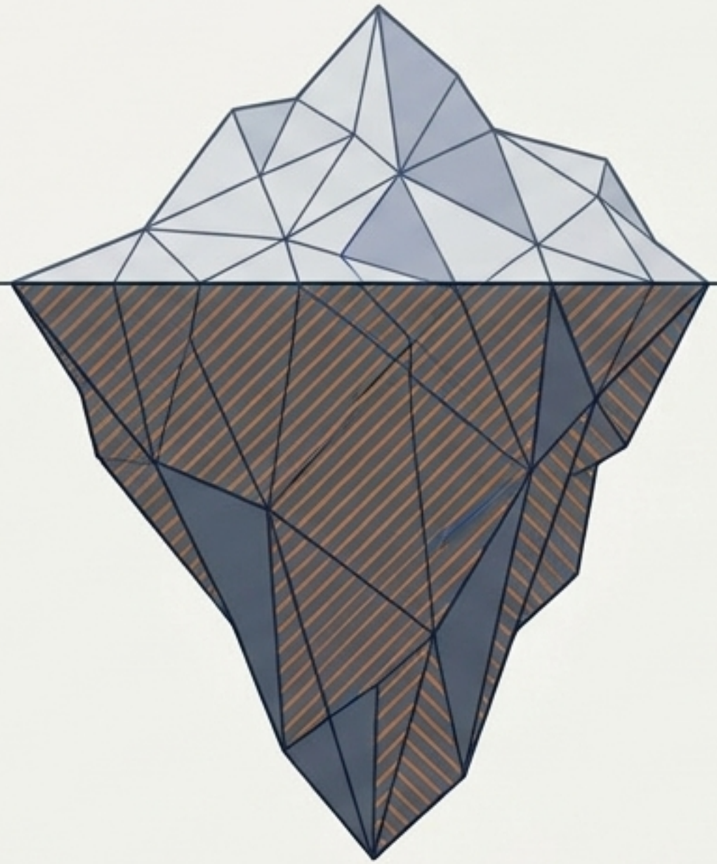
When a system defect causes 100 customer transactions to fail, the cost is exponentially higher than the developer's fixing time.

Direct Costs

Developer fixing time, repeated testing, data correction.

Hidden Operational Costs

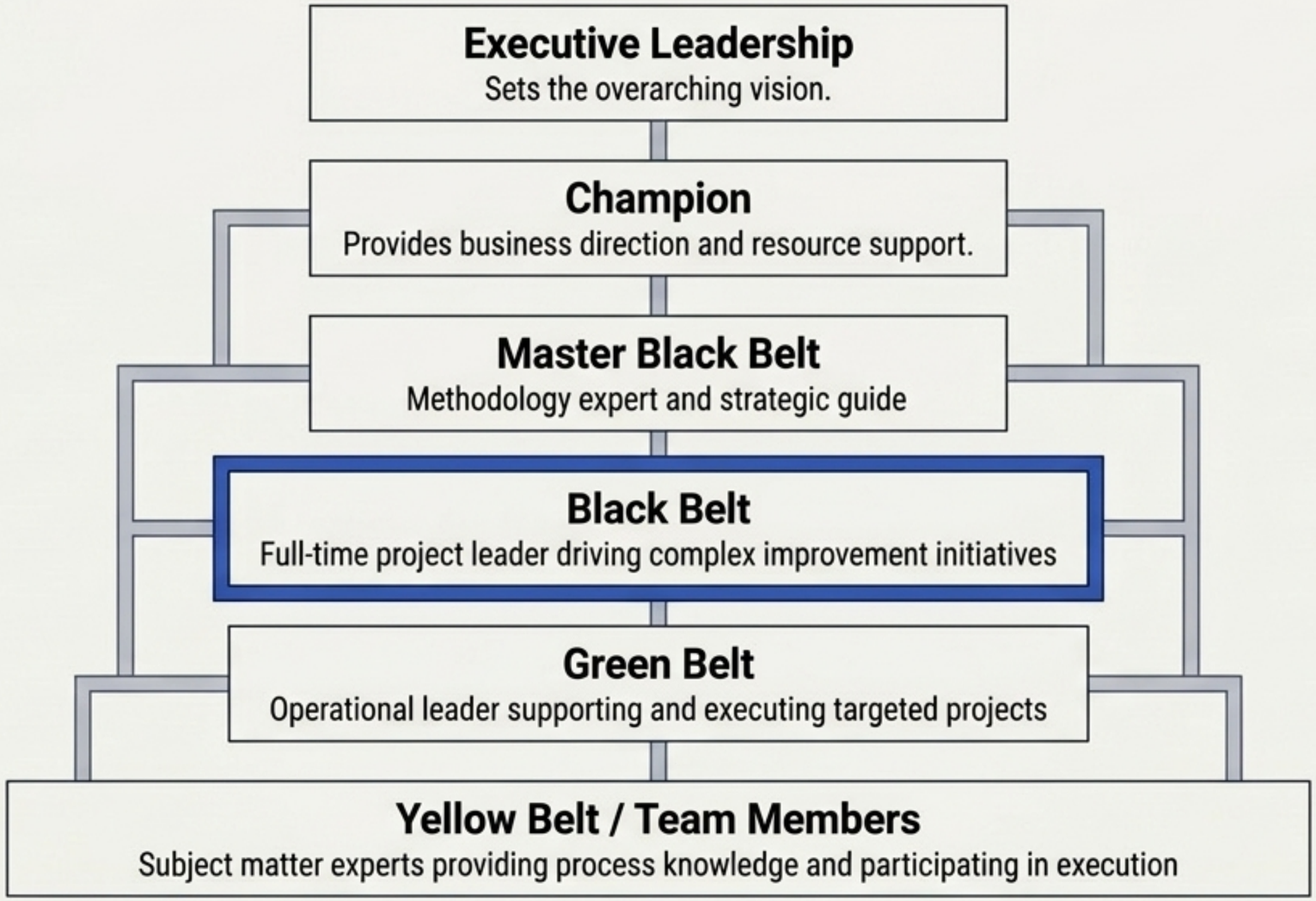
Support tickets flooded, business operation downtime, customer complaints, refunds, customer churn, and total revenue impact.



The COPQ Iceberg Metaphor

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System Operators: The Lean Six Sigma Organization



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The Synthesis: Lean Six Sigma 4.0

The Ecosystem Equation

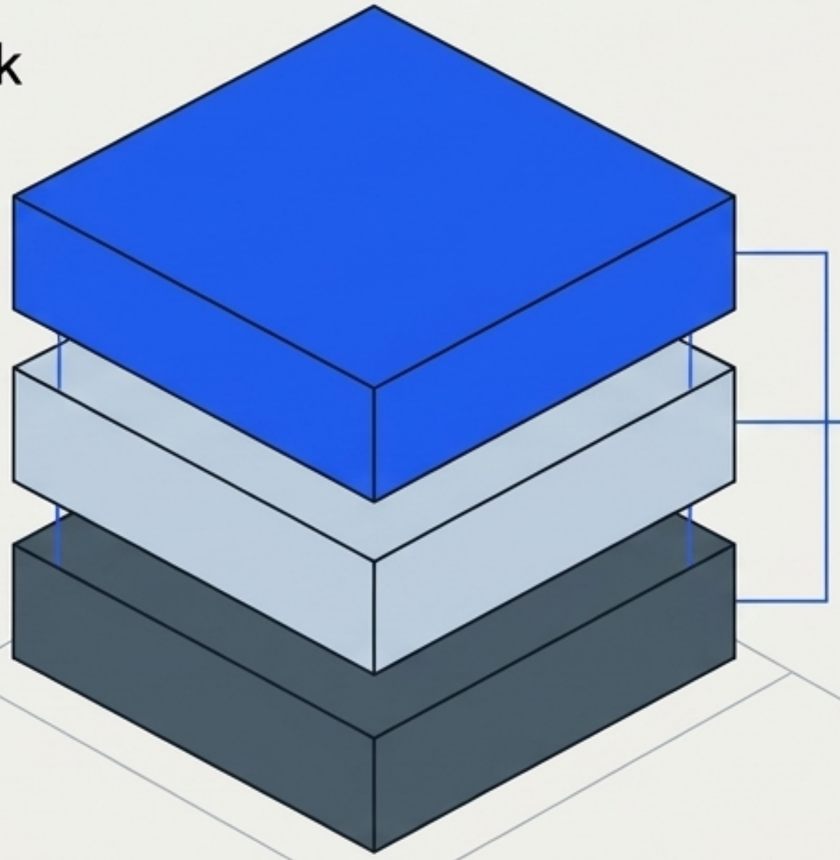
(Lean's Waste Removal) + (Six Sigma's Variation Reduction) × (AI & Data's Speed & Scale) = LSS 4.0

The Architecture Stack

AI & Automation Layer
Analytics, Machine Learning, and RPA Insight and Continuous Improvement.

Data Layer
Digital Data & Process Mining capturing system reality in real-time.

Process Layer
Traditional foundation mapping work and identifying waste.

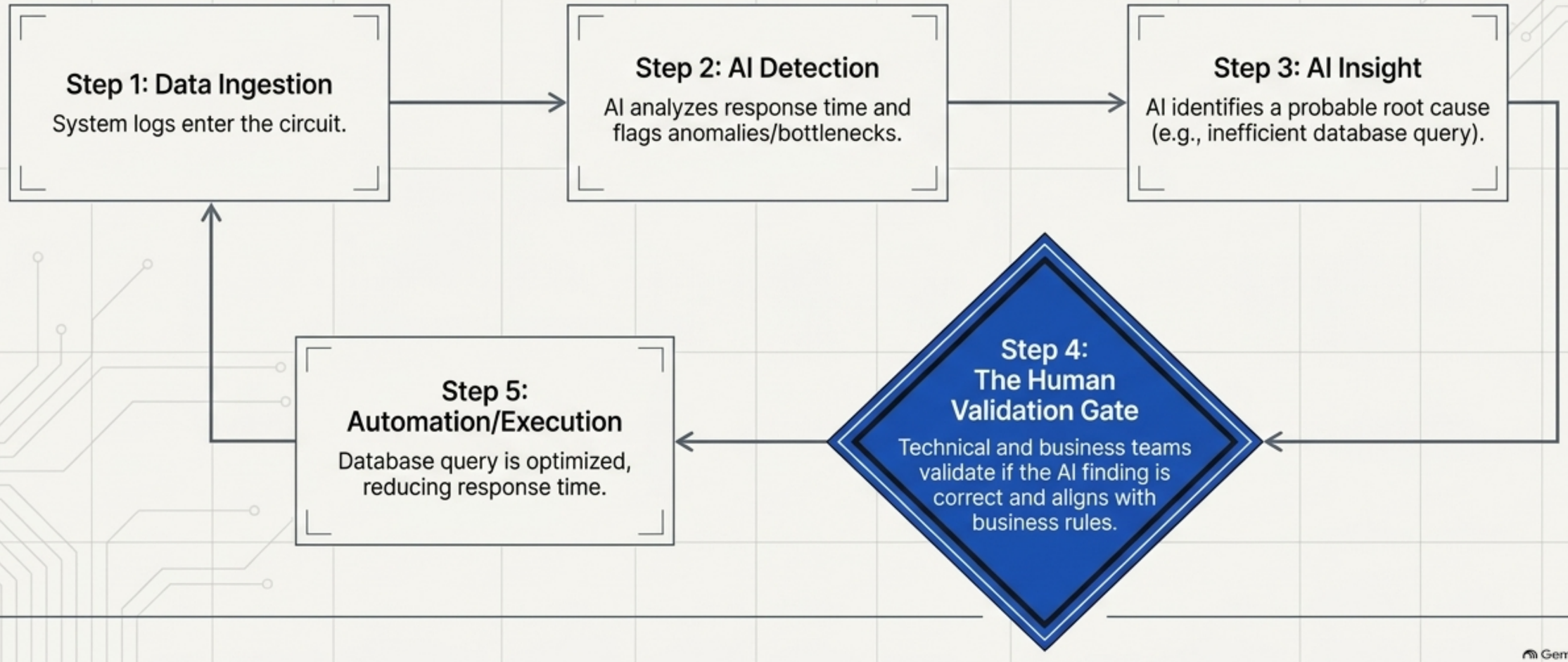


Technologies powering the stack: IoT, Cloud, RPA, Digital Dashboards, ML.

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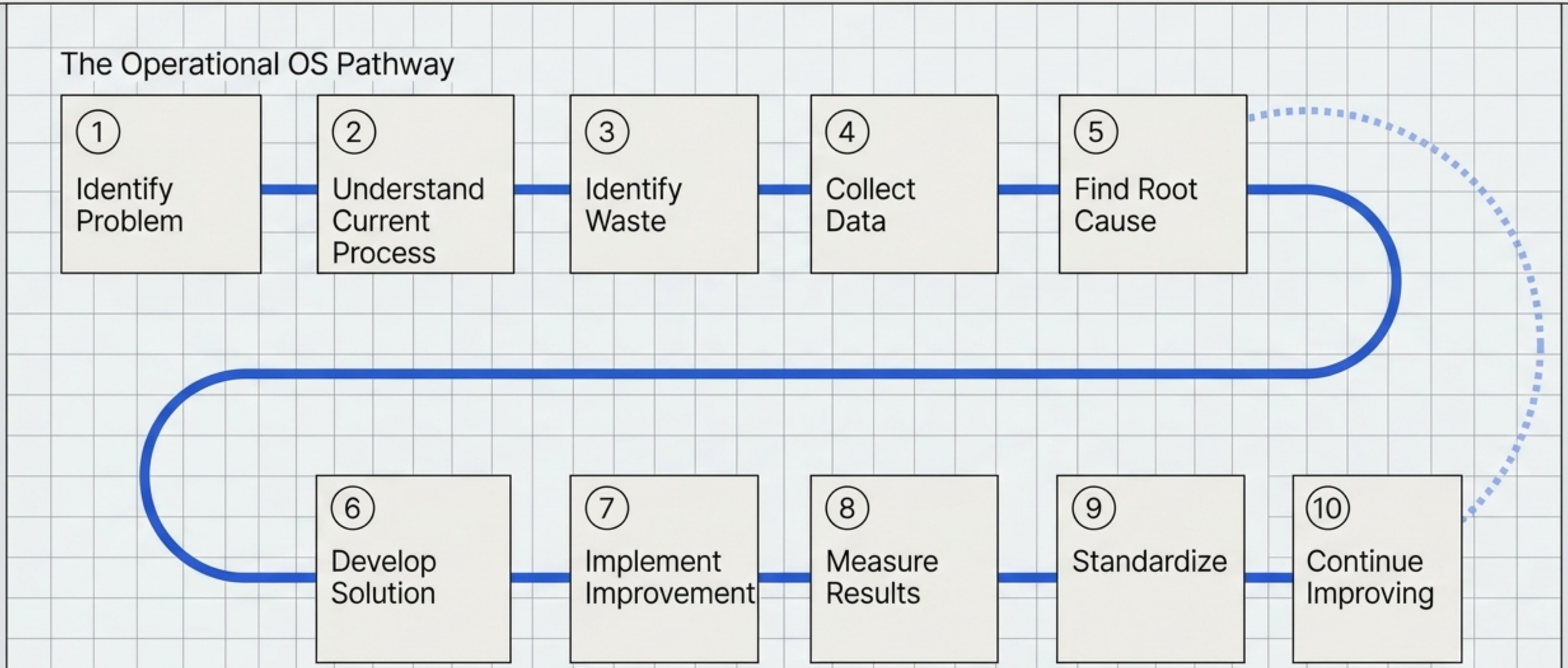
The Human-in-the-Loop AI Circuit

AI supports human decision-making and accelerates analysis; it should **never** replace business validation.



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The Kaizen IT Playbook: 10 Steps to Execution



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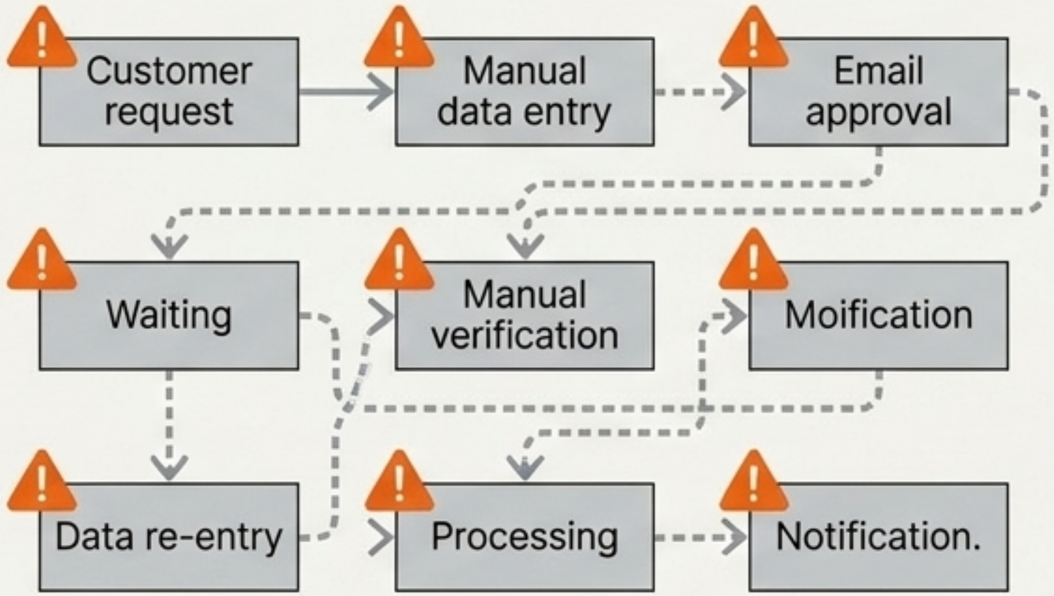
Case Study: Digital Process Transformation

Problem: Customer request processing takes 5 days due to manual workflow, duplicate data entry, and frequent follow-up emails.

Before vs. After Automation Matrix

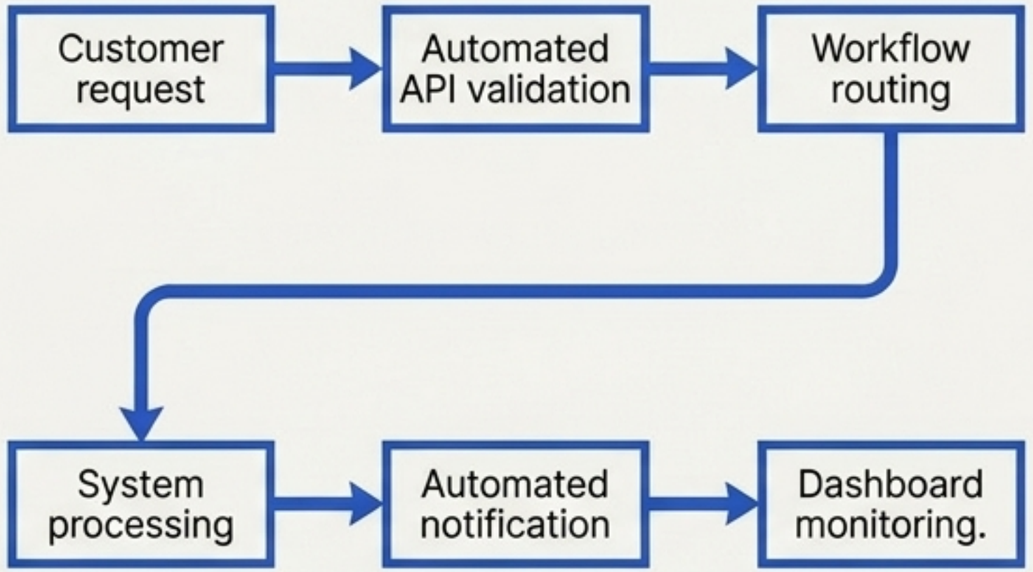
LSS 4.0 Future

Before



Result: 5 Days.

After



Target Result: 2 Days.

Tangible outcomes: Less waiting, zero duplicate entry, faster processing, complete system visibility.

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The LSS 4.0 Command Dashboard: Measuring Impact

Time (Speed)

- Lead Time
- Cycle Time
- Processing Time

Quality (Precision)

- Defect Rate
- Error Rate
- Rework Rate
- First-Time-Right %

Customer (Experience)

- Customer Satisfaction (CSAT)
- Complaint Rate
- Response Time

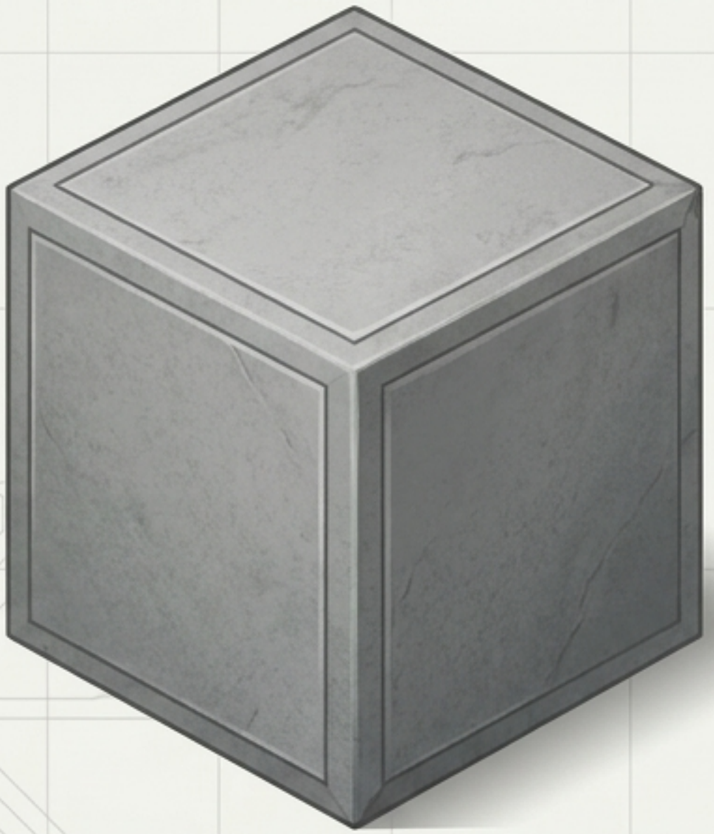
Business (Value)

- Cost Reduction
- Productivity Throughput
- Revenue Impact

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The OS Update: Shifting to a Continuous Improvement Mindset

The Legacy Mindset (Static)



This is how we have always done it.

The LSS 4.0 Mindset (Dynamic)



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The Lean Practitioner's Mandate

Lean Six Sigma is not just a set of tools. It is a way of thinking. Your role is not simply to find problems, but to understand processes, use data to find root causes, and ensure improvements are sustainable.

Lean:	Relentlessly remove waste.
Six Sigma:	Use data to reduce variation and defects.
Data & AI:	Automate analysis to make faster, better decisions.
Kaizen:	Facilitate continuous, incremental improvement.

Better Quality + Lower Cost + Faster Process = Ultimate Customer Value.

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